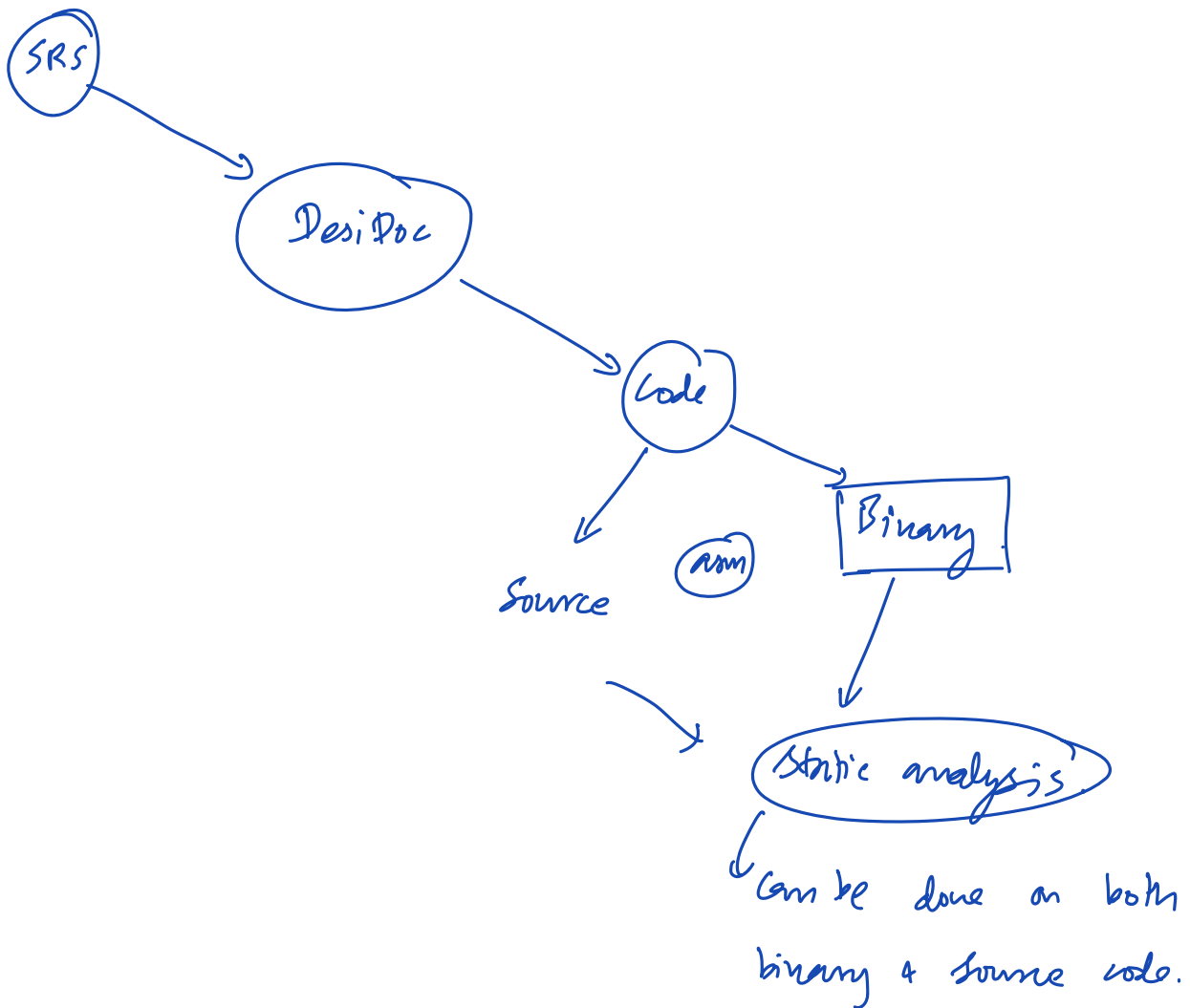


final executable → build → assembly

source . code $\xrightarrow{B/C}$



1 Gen language : Machine Code

2G : ASM

3G : High-level: C, C++, java

$$P = L + C + D/S$$

4G : SQL, ...

5G : Prologue, mercury

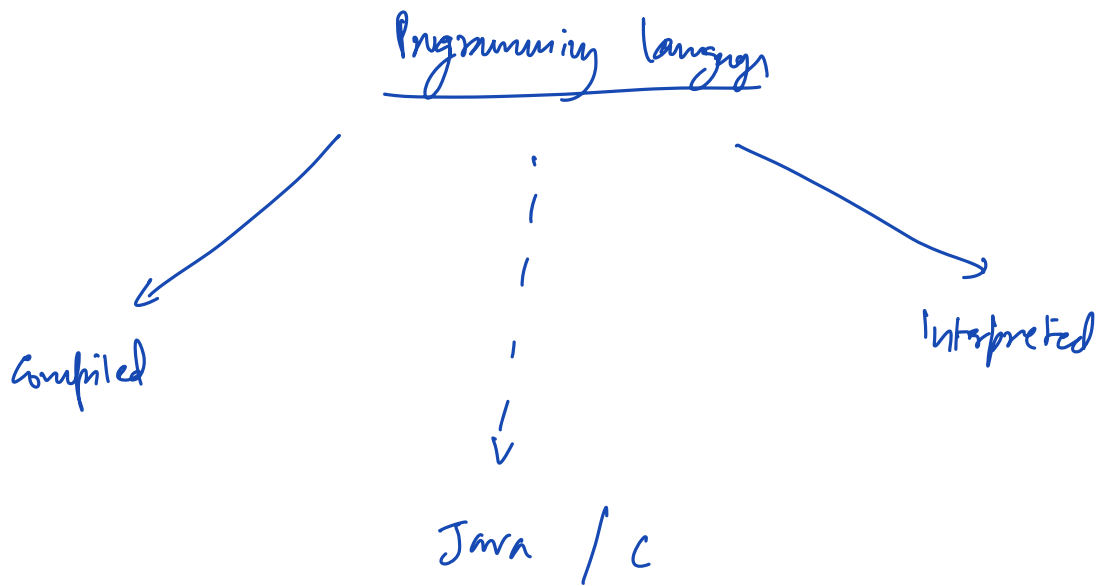
6G :

$$P = L \cup C \cup D/S$$

With 4G, you need to write all 3.

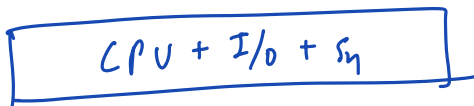
5G : control is taken care of. → being provisioned by the runtime of the language.

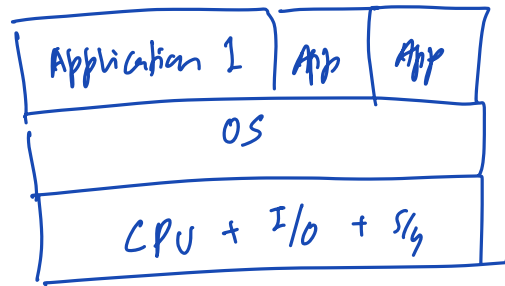
$$P_{5G} = L \cup D/S$$



Java: platform independent

- ① Symbols
- ② compiler
- ③ target code
- ④





Q) when we write a source code $\rightarrow$ two types of compilation can be done $\rightarrow$ static and dynamic compilation

$\uparrow$
This was done yesterday.

dynamic $\rightarrow$ we don't ship the entire compilation
we only ship

static compilation

If you don't know about your destination for a trip, you

baggage increases. $\Rightarrow$ static compilation

ships the libraries with the binary.

next week $\rightarrow$ worksheet $\rightarrow$ submit on last session in the week.

Java

Common Language Runtime
(.net)

Java : file.java $\xrightarrow{\text{compile}}$ Java C $\longrightarrow$ bytecode

① $\rightarrow$ ELF (only on linux)
 $\rightarrow$ EXE (only windows)



Java $\longrightarrow$ Byte code

Install : ① Up2date

② Spin

Bin

biname
runtime
environment

Bytecode $\rightarrow$ (platform independent)

Java Runtime Environment

JVM $\rightarrow$ runs bytecode
(Java virtual machine)

program $\rightarrow$ simple mathematical object



file language

program $\rightarrow$ transition system

$\langle S, \Delta, ISS \rangle$

$\downarrow$
(intermediate stable state)

```
int main() {
```

```
    int i;
```

```
    double d;
```

```
    char c;
```

```
    statement 1
```

$\langle i, d, c \rangle$

$i \in \mathbb{Z}$

$d \in \mathbb{O}$

$c \in$

```
    return 0;
```

```
}
```

Embedded systems

hybrid

- Realtime

